

Supplementary material: From Candidate Scores to Plant Recovery in Finite Crop–Weed Review Queues

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S1. Analysis chronology and truth-access governance

WeedsGalore and CWFID serve different evidential roles. WeedsGalore supports internal development on its official train, validation, and test split. Several extended WeedsGalore analyses were conducted after the initial official-test archive existed and are interpreted as within-dataset engineering analyses. CWFID was separately frozen and opened once, without external fitting, threshold selection, or policy search. Table S1 records this chronology. Test truth was unavailable to every queue-forming stage and entered only the offline scorer.

The CWFID receipt records one opening against a maximum of one. The external protocol, dataset contract, model list, preprocessing, component definition, matching thresholds, fixed $K = 20$, ranking rule, endpoints, and 10,000-replicate image bootstrap were fixed before that opening. The later zero-candidate consolidation read only the already written per-image tables and did not reopen images or annotations.

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Table 1: Chronology and data roles. Times are local time (UTC+08:00) where an electronic freeze or open receipt exists.

Event	Recorded time	Analysis role	Truth access
Core WeedsGalore archive	Earlier archived analysis	Internal development and official-split scoring	Train for fitting; validation for selection; test for offline scoring
Target-semantic and queue extensions	10–12 Aug 2026	Within-dataset engineering analysis	Validation selected epochs and policies; test remained scoring-only
CWFID protocol freeze	12 Aug 2026, 18:46:11	Locked cross-dataset design	Public metadata only; images and annotations unopened
Strong-baseline freeze	12 Aug 2026, 18:58:00	Model-family extension with validation-only selection	No CWFID image or annotation access
Matched-burden freeze	12 Aug 2026, 19:08–19:17	Paired image-level audit	Existing test outputs only; no fitting on test truth
CWFID opening	12 Aug 2026, 19:54:27	Single locked external evaluation	Annotations used only after all model outputs were fixed
Two-operator review	12 Aug 2026	Technical review of one frozen queue	Operators saw RGB views only; mask truth and model metadata were hidden

S2. Public-data lineage, split denominators, and annotation audit

All imagery used in the reported experiments is public. SugarBeets2016 supplied source-specialist masks; WeedsGalore supplied the internal development study; CWFID supplied the locked cross-dataset test. Public images and public model weights remain at their upstream repositories. The reproducibility release records source URLs, access dates, licences, revisions, byte counts, and hashes.

The public release describes one maize field, 48 recorded locations, and an 8/2/2 spatial allocation. The machine-readable release provides official

Table 2: Official WeedsGalore split by acquisition date. Instance counts are obtained by enumerating the released rasters.

Date	Train tiles	Validation tiles	Test tiles	Test crop	Test weed
25 May 2023	32	8	8	122	392
30 May 2023	32	8	8	55	440
6 June 2023	32	8	8	68	726
15 June 2023	8	2	2	11	154
Total	104	26	26	256	1,712

split labels and capture indices, but not a verified mapping from each capture index to an independently sampled physical field unit. We therefore retain the official split and acquisition dates and do not treat capture-index groups as independent farms. Across all 156 rasters, enumeration yielded 2,168 crop and 10,029 weed instances. The annotation audit found no role disagreement between semantic and instance rasters and no excluded tile identifier.

CWFID v1.0 was fixed at upstream commit `176612f3506c0ce70d85518cb82ddd938a6206c8`. All 242 expected upstream blobs passed byte-count and Git-blob SHA-1 verification. The 60 RGB–annotation pairs passed dimension, colour-semantics, and pairing audits. After the frozen resize and padding transformation, the primary truth rule produced 331 weed components of at least 16 pixels. These connected components are evaluation proxies; they are not persistent plant identifiers.

S3. Supervision ledger and model construction

The source foreground and semantic specialists were ResNet-18 U-Net models initialised from public ImageNet-1K weights. They used ImageNet RGB normalisation, AdamW, bfloat16 autocasting, gradient clipping at 5, and deterministic 128×128 patches. The foreground model used 32,768 patches, binary cross-entropy with positive weight 30 plus soft Dice, and 10 epochs. The semantic model used 16,384 role-balanced patches, weighted cross-entropy with weights (0.2, 1, 1) plus plant-role Dice, and 8 epochs. On the 901-image source diagnostic split, the foreground scan reached maximum pixel IoU 0.3821 at threshold 0.99. The semantic specialist reached three-class mean IoU 0.4425; its independently selected role operating point had balanced accuracy 0.7224, crop recall 0.5524, and weed recall 0.8924.

Table 3: Complete supervision ledger. “Read by selection” distinguishes labels present in a public dataset from labels actually used to choose an operating point.

Information	Available amount	Read by selection	Purpose
SugarBeets pixel masks	7,050 train; 901 validation	Source training and source diagnostics	Foreground and crop-weed specialists
WeedsGalore train masks	104 tiles	Proposal setting and target-model fitting	Dense loss, component generation, candidate labels
WeedsGalore validation masks	26 tiles	Epoch, component, regularisation, and ranking-policy selection	Model and queue selection
WeedsGalore test masks	26 tiles	None	Offline proposal, queue, IoU, matching, and crop-overlap scoring
Restricted candidate-label experiment	300 sampled train candidates per draw	All 2,436 labelled validation candidates remain available	Training-label acquisition comparison, not end-to-end 300-label learning
CWFID annotations	60 images	None	Single locked external score
Operator responses	518 regions per operator	None	Agreement and reviewability analysis; no model update

Semantic models trained on random 512×512 crops and were evaluated at 600×600 . Weight decay was 10^{-4} . Candidate thresholds were 0.30, 0.50, and 0.70; component areas were 16, 32, and 64 pixels. Mask R-CNN used a 600-pixel minimum and maximum side, mask threshold 0.50, detection-score floor 0.05, momentum 0.9, and weight decay 5×10^{-4} . Its queue placed predicted weed instances before other detections and then sorted by detection confidence.

Table 4: Target-trained model recipes. Public initialisation weights were stored by hash before training.

Model	Initialisation	Loss	Training	Validation selection
ResNet-18 U-Net	ImageNet-1K	weighted CE + crop/weed Dice	AdamW, 3×10^{-4} ; 40 epochs; 5 seeds	epoch, threshold, area, rank score
DeepLabV3+ ResNet-50	ImageNet-1K	weighted CE + crop/weed Dice	AdamW, 3×10^{-4} ; 30 epochs; 3 seeds	epoch, threshold, area
Mask R-CNN- ResNet-50-FPN	COCO V1	standard detection/mask losses	SGD, 5×10^{-3} ; 20 epochs; 3 seeds	epoch

S4. Proposal definitions, operating-point selection, and matching sensitivity

For truth instance I , candidate C , all labelled plant pixels L , and pixels of agricultural role R , the offline scorer records

$$q_I = \frac{|I \cap C|}{|I|}, \quad q_L = \frac{|C \cap L|}{|C|}, \quad q_R = \frac{|C \cap R|}{|C \cap L|}.$$

Spatial support requires $q_I \geq 0.50$. Role-qualified support additionally requires $q_L \geq 0.50$, $q_R \geq 0.90$, and a role match. One-to-one recovery uses a maximum-cardinality bipartite matching over eligible candidate–instance edges. Candidate identifiers and then truth identifiers define the deterministic lexicographic tie order.

The selected proposal setting was threshold 0.70 and area 32. The deterministic order was higher training qualified recall, lower mean burden, higher threshold, and larger minimum area. The original comparison setting was threshold 0.95 and area 32.

Raw candidate composition changed from 453 weed-qualified, 148 crop-qualified, and 201 ambiguous candidates (802 total) to 464, 111, and 1,776 (2,351 total). The original burden had median 28.5, interquartile range 19.25–43.75, maximum 62, and seven tiles with fewer than 20 candidates. The commissioned burden had median 89, interquartile range 60–114.25, maximum 183, and no tile with fewer than 20 candidates.

Table 5: Official-training proposal grid. Feasible settings have mean burden no greater than 100 candidates per tile.

Threshold / minimum area (px)	Candidates/tile	Spatial weed recall	Qualified weed recall	Feasible
0.30 / 8	281.35	0.7135	0.2406	No
0.30 / 32	142.04	0.7000	0.2297	No
0.50 / 8	259.49	0.6325	0.2822	No
0.50 / 32	123.52	0.6190	0.2707	No
0.70 / 8	207.48	0.5416	0.2979	No
0.70 / 32	95.07	0.5271	0.2856	Yes
0.80 / 8	168.63	0.4806	0.2959	No
0.80 / 32	76.99	0.4672	0.2835	Yes
0.90 / 8	115.50	0.3703	0.2421	No
0.90 / 32	52.38	0.3587	0.2317	Yes
0.95 / 8	75.71	0.2452	0.1685	Yes
0.95 / 32	36.83	0.2366	0.1605	Yes

Table 6: Official-test proposal layers by date. Recall numerators count weed instances.

Date	Setting	Weed n	Spatial support	Qualified support	Candidates/tile
25 May	Original	392	2 (0.0051)	2 (0.0051)	13.88
25 May	Commissioned	392	51 (0.1301)	13 (0.0332)	102.75
30 May	Original	440	106 (0.2409)	84 (0.1909)	35.75
30 May	Commissioned	440	263 (0.5977)	121 (0.2750)	119.38
6 June	Original	726	328 (0.4518)	237 (0.3264)	45.13
6 June	Commissioned	726	566 (0.7796)	369 (0.5083)	64.63
15 June	Original	154	73 (0.4740)	7 (0.0455)	22.00
15 June	Commissioned	154	98 (0.6364)	15 (0.0974)	28.50
Pooled	Original	1,712	509 (0.2973)	330 (0.1928)	30.85
Pooled	Commissioned	1,712	978 (0.5713)	518 (0.3026)	90.42

Table 7: Split/merge sensitivity across all 156 WeedsGalore tiles. A merge is a candidate overlapping at least two truth instances at the listed instance-coverage threshold.

Setting	Coverage threshold	Split instances	Merge candidates
Original	0.10	224	735
Original	0.25	40	640
Original	0.50	0	473
Commissioned	0.10	154	1,136
Commissioned	0.25	26	1,065
Commissioned	0.50	0	938

The complete matching-contract grid crosses instance coverage $\{0.25, 0.50, 0.75\}$, candidate labelled coverage $\{0.25, 0.50, 0.75\}$, and same-role purity $\{0.50, 0.75, 0.90\}$. For the masked-DINOv2 $K = 20$ queue, many-to-one qualified recall ranges from 0.0485 to 0.2886 and one-to-one recall from 0.0374 to 0.1881. At the primary 0.50/0.50/0.90 contract, these are 319/1,712 (0.1863) and 230/1,712 (0.1343). Under IoU-only matching at thresholds 0.25/0.50/0.75, many-to-one recall is 325/1,712, 163/1,712, and 24/1,712; one-to-one recall is 307/1,712, 163/1,712, and 24/1,712.

Table 8: Crop-overlap tile frequency for frozen-DINOv2 queues at $K = 20$. The first column requires any positive crop overlap.

Ranker	Any > 0	Crop fraction ≥ 0.01	≥ 0.10	≥ 0.50
Full-region DINOv2	10/26	10/26	10/26	9/26
Component-masked DINOv2	11/26	11/26	11/26	9/26

S5. Matched review burden and proposal commissioning

The matched comparison applied one role-plus-local logistic ranker to both proposal pools without changing its weights. Logistic regularisation was selected on validation; $C = 100$ was selected from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. For each test tile, both proposal settings used the same $k_i = \min(20, n_i^{\text{original}}, n_i^{\text{commissioned}})$, yielding 471 selected candidates from each pool.

Table 9: Paired common-capacity results and 95% image-cluster bootstrap intervals.

Setting	Precision	Spatial recall	Qualified recall	One-to-one recall	Crop-exposed tiles
Original	0.5860 [0.4430, 0.7152]	0.1875 [0.1166, 0.2747]	0.1197 [0.0826, 0.1583]	0.0970 [0.0698, 0.1236]	0.8462 [0.6923, 0.9615]
Commissioned	0.5180 [0.3831, 0.6464]	0.2693 [0.1816, 0.3752]	0.1571 [0.1077, 0.2085]	0.1157 [0.0848, 0.1453]	0.8846 [0.7308, 1.0000]

Global equal- N ranking also selected 471 candidates from each pool. One-to-one recall was 0.0970 for the original pool and 0.1174 for the commissioned pool. The paired common- K AURBC values were 0.0470 and 0.0578; global equal- N AURBC values were 0.0598 and 0.0750. These analyses isolate proposal support from the 49-candidate difference that occurs when a nominal $K = 20$ is applied independently to each pool.

Table 10: Commissioned-minus-original paired effects under common per-tile capacity.

Metric	Difference	95% image-bootstrap interval
Candidate precision	-0.0679	$[-0.1384, 0.0064]$
Spatial recall	0.0818	$[0.0560, 0.1098]$
Role-qualified recall	0.0374	$[0.0166, 0.0580]$
One-to-one recall	0.0187	$[0.0032, 0.0347]$
Crop-exposed tile frequency	0.0385	$[0, 0.1154]$

Table 11: Candidate-level discrimination on the complete 2,351-candidate WeedsGalore test pool. AUC and AP are diagnostics rather than plant-level endpoints.

Model and fitting support	AUC	AP
Source weed score, all candidates	0.4947	0.2877
Eligible-only role+local model, evaluated on all candidates	0.4420	0.1640
All-candidate role+local model	0.9049	0.6960
All-candidate geometry model	0.8154	0.5418
Eligible-only frozen DINOv2, evaluated on all candidates	0.6321	0.3816
All-candidate frozen DINOv2	0.9630	0.8684
All-candidate component-masked DINOv2	0.9665	0.8813

S6. Candidate-ranking diagnostics, ceilings, and restricted label draws

Component masking changed AUC by 0.0035 and AP by 0.0129 in one paired run. This small diagnostic difference is not used as a method claim. At $K = 20$, the masked-DINOv2 queue selected 520 regions, of which 278 were role-qualified weed candidates. Many-to-one spatial and qualified recall were 459/1,712 (0.2681) and 319/1,712 (0.1863); one-to-one qualified recall was 230/1,712 (0.1343). The role-qualified proposal ceiling was 518/1,712 (0.3026), the exact unconstrained oracle was 426/1,712 (0.2488), and 1,000 random rankings had 2.5th/97.5th percentiles 0.077/0.104 for qualified recall at $K = 20$.

The restricted-label analysis sampled 300 training candidates per draw but retained all 2,436 labelled validation candidates for regularisation selection. It therefore measures a restricted training-candidate reveal, not an end-to-end 300-label system.

Paired last-block fine-tuning on the ten coreset draws changed AUC by

Table 12: Candidate discrimination by acquisition date for the masked-DINOv2 ranker. Class counts accompany AUC and AP.

Date	Candidates	Weed positive	Other	AUC	AP
25 May	822	15	807	0.8798	0.2131
30 May	955	132	823	0.9535	0.7492
6 June	517	279	238	0.9381	0.9402
15 June	57	38	19	0.7382	0.8577
Pooled	2,351	464	1,887	0.9605	0.8617
Date macro	—	—	—	0.8774	0.6901

Table 13: Mean official-test results over ten draws of 300 training-candidate labels plus the fully labelled validation candidate set.

Training-candidate selection	AUC	AP	$K = 20$ precision	Qualified recall
Prospective uniform	0.9549	0.8544	0.5242	0.1724
Prospective DINOv2 coreset	0.9543	0.8534	0.5217	0.1717
Retrospective class-stratified reference	0.9547	0.8502	0.5231	0.1725

−0.0396 on average (empirical 2.5th/97.5th percentiles −0.0525/ − 0.0170), candidate precision by −0.0373, and qualified recall by −0.0115. The complete per-draw differences and selected candidate identifiers are retained in the archive.

S7. Strong baselines, semantic ranking, and acquisition-date variation

For the five U-Net candidate pools, validation chose mean weed probability in four seeds and conditional weed probability in one. The selected policies had mean $\pm$ sample SD precision 0.7440 ± 0.0185 , spatial recall 0.3391 ± 0.0249 , qualified recall 0.2873 ± 0.0082 , one-to-one recall 0.2117 ± 0.0047 , and crop-exposed tile frequency 0.2846 ± 0.0516 . Their candidate AUC and AP were 0.9021 ± 0.0135 and 0.9417 ± 0.0117 and remain diagnostic.

Standard semantic IoU complements the queue endpoint. Across five U-Net seeds, background/crop/weed IoU were 0.9783 ± 0.0013 , 0.6884 ± 0.0279 , and 0.7197 ± 0.0120 ; three-class mean IoU was 0.7955 ± 0.0127 . Across three DeepLabV3+ seeds, corresponding values were 0.9755 ± 0.0013 , 0.6795 ± 0.0063 , 0.7021 ± 0.0121 , and 0.7857 ± 0.0030 .

Table 14: Per-seed $K = 20$ queue results. Precision and recalls use the fixed role-qualified matching contract; crop is the fraction of 26 tiles with any crop overlap.

Model	Seed	Precision	Spatial recall	Qualified recall	One-to-one recall	Crop
U-Net	20260810	0.7112	0.2704	0.2494	0.2062	0.2692
U-Net	20260811	0.7248	0.2634	0.2477	0.1992	0.3077
U-Net	20260812	0.6879	0.2523	0.2266	0.2033	0.2308
U-Net	20260813	0.7569	0.3008	0.2833	0.2097	0.2308
U-Net	20260814	0.7035	0.2471	0.2272	0.1951	0.1923
DeepLabV3+	20260820	0.6848	0.2763	0.2436	0.1945	0.2692
DeepLabV3+	20260821	0.6863	0.2588	0.2319	0.1887	0.2308
DeepLabV3+	20260822	0.6784	0.2780	0.2506	0.1939	0.2308
Mask R-CNN	20260820	0.8000	0.2453	0.2313	0.2202	0.2308
Mask R-CNN	20260821	0.7942	0.2371	0.2255	0.2132	0.1538
Mask R-CNN	20260822	0.7731	0.2418	0.2208	0.2144	0.3077

Table 15: Validation-selected checkpoints and component policies. Mask R-CNN uses its fixed score floor and therefore has no component threshold or minimum area.

Model	Seed	Epoch	Foreground threshold / area	Ranking policy
U-Net	20260810	38	0.50 / 16	Mean weed probability
U-Net	20260811	36	0.70 / 16	Mean weed probability
U-Net	20260812	40	0.30 / 16	Mean weed probability
U-Net	20260813	39	0.50 / 16	Conditional weed probability
U-Net	20260814	31	0.50 / 16	Mean weed probability
DeepLabV3+	20260820	24	0.30 / 16	Conditional weed probability
DeepLabV3+	20260821	28	0.50 / 16	Conditional weed probability
DeepLabV3+	20260822	24	0.50 / 16	Conditional weed probability
Mask R-CNN	20260820	16	–	Weed class, then confidence
Mask R-CNN	20260821	16	–	Weed class, then confidence
Mask R-CNN	20260822	18	–	Weed class, then confidence

Table 16: Paired model-minus-U-Net contrasts from 10,000 image-cluster bootstrap resamples. The tile metric is first averaged across training seeds within each model.

Contrast	Metric	Difference	95% interval
DeepLabV3+ minus U-Net	Precision	−0.0337	[−0.0705, 0.0013]
DeepLabV3+ minus U-Net	Spatial recall	0.0042	[−0.0090, 0.0170]
DeepLabV3+ minus U-Net	One-to-one recall	−0.0103	[−0.0216, −0.0002]
Mask R-CNN minus U-Net	Precision	0.0723	[0.0231, 0.1370]
Mask R-CNN minus U-Net	Spatial recall	−0.0254	[−0.0485, −0.0015]
Mask R-CNN minus U-Net	One-to-one recall	0.0132	[0.0015, 0.0284]

Table 17: U-Net one-to-one recall by acquisition date. Values are mean $\pm$ sample SD across five seeds.

Date	Tiles	Weed instances	One-to-one recall
25 May	8	392	0.1378 ± 0.0236
30 May	8	440	0.2864 ± 0.0146
6 June	8	726	0.2061 ± 0.0071
15 June	2	154	0.1130 ± 0.0098

Mean candidate precision for the same dates was 0.4035, 0.8163, 0.9513, and 0.5650. The 15 June stratum contains only two tiles; its date-specific value is therefore reported with its denominator and is not treated as an independent field estimate.

S8. Two-operator technical review

One frozen 518-region target-semantic queue was presented to two operators. Every item showed the source RGB tile, a highlighted region, and a local crop. Date, scene identity, candidate identifier, score, rank, model and policy names, and mask-derived truth were hidden. Both operators completed all 518 items independently, did not exchange labels or rationales, and did not inspect one another’s responses. The analysis excluded the unused timing and submission-time fields.

Table 18: Two-operator categorical agreement on 518 regions.

Field	Exact agreements	Agreement	Cohen’s κ
Primary content	491	0.9479	0.9095
Weed reviewability	488	0.9421	0.8930
Crop presence	490	0.9459	0.8492
Localisation usability	487	0.9402	0.9049

For weed reviewability, both operators answered yes for 301 regions and no for 143; 74 regions were disputed or contained at least one uncertain response. A 10,000-replicate scene-cluster bootstrap gave a 95% percentile interval of 0.9228–0.9611 for exact agreement and 0.8474–0.9249 for κ . Of the 301 consensus-positive regions, 249 satisfied the mask-derived role-qualified weed contract, a correspondence fraction of 0.8272. The mask contract identified

355 qualified weed candidates in the same queue. These values compare two representations of usefulness—human reviewability and geometric mask qualification—without using operator responses to tune the system. The sample supports inference only to the two fixed operators and presented queue.

S9. Locked CWFID evaluation

The frozen external pipeline scaled the longer image side to 600 pixels, reflection-padded the shorter side, and applied the training normalisation. Annotations used nearest-neighbour resize and zero padding. Each of five existing U-Net checkpoints retained its WeedsGalore-validation-selected epoch, foreground threshold, minimum component area, and mean/conditional probability score. CWFID labels were not used for fitting, calibration, threshold selection, or allocation.

Table 19: Locked CWFID result. All five checkpoints produced the same values.

Seeds	Images	Weed components	Candidates	Accepted	One-to-one recall	Crop exposure
5	60	331	0	0	0/331 (0)	0/60 (0)

The image-cluster bootstrap interval for the primary endpoint was 0–0. Background IoU was 0.9450, while crop and weed IoU were both 0, yielding three-class mean IoU 0.3150. Every checkpoint predicted background for every pixel at its frozen operating point. The original evaluator successfully wrote the 300 per-image seed rows but stopped when its table writer rejected an empty candidate table. A versioned consolidator then computed the zero-candidate result from those existing rows only. It did not reopen external data, repeat inference, or change a threshold. Both the failed evaluator record and the completed consolidation are retained.

S10. Exploratory allocation and development diagnostics

The principal queue capacity is fixed $K = 20$. A bounded 5–50 candidates-per-tile allocation was examined after the initial WeedsGalore test archive and was not selected by validation in the five-seed study. It is retained as a hypothesis-generating allocation analysis and is absent from the abstract and principal conclusions.

Table 20: Exploratory allocation for one masked-DINOv2 queue. All policies except the validation-threshold row accept 520 candidates.

Policy	Accepted	Precision	Spatial recall	Qualified recall	One-to-one recall	Crop-exposed tiles
Fixed 20/tile	520	275/520	390/1,712	324/1,712	227/1,712	13/26
Bounded 5–50/tile	520	382/520	533/1,712	453/1,712	315/1,712	14/26
Global top- N diagnostic	520	388/520	547/1,712	464/1,712	320/1,712	12/26
Validation score threshold	426	348/426	485/1,712	424/1,712	290/1,712	8/26

The bounded policy assigned 54.8% of its reviews to 6 June and had allocation Gini 0.3817; the global top- N diagnostic assigned 56.4% to 6 June and had Gini 0.4220. Their worst-date one-to-one recall was 0.0153, below 0.0255 for fixed $K = 20$. The apparent pooled gain therefore arose with increased allocation concentration and did not constitute locked transfer evidence.

Table 21: Development diagnostics retained with their observed direction.

Analysis	Result	Interpretation
Nested Platt calibration	Date-macro Brier 0.2112 before and 0.2776 after; NLL 0.6338 before and 0.8239 after	Date transfer increased both calibration losses
Empirical threshold transfer	Precision 0.1317; empty response 0.7448; crop frequency 0.1042	A training-date threshold transferred sparsely
Cost-sensitive ranker at $K = 20$	AUC 0.7301; AP 0.7272; precision 0.4327; instance recall 0.0842	Crop weighting changed recovery and crop exposure in opposite directions
Last-block DINOv2 fine-tuning	Mean paired AUC change -0.0396 ; qualified-recall change -0.0115	The archived small-label recipe did not improve the frozen representation

S11. Reproducibility, environment, and preserved failures

The public release linked in the main manuscript contains frozen configurations, model-acquisition records, environment specifications, training and evaluation source, checkpoint hashes, candidate and matching outputs, source tables for every figure, failure records, and clean-room replay commands. Public dataset imagery and public model weights are acquired from upstream sources and are not redistributed when upstream terms do not permit it.

The principal training environment used Python 3.10.12, PyTorch 2.7.1 with CUDA 12.6, torchvision 0.22.1, and segmentation-models-pytorch 0.5.0. Training runs used NVIDIA H200 accelerators assigned to physical devices 6 and 7; one accelerator was used by each training process. CPU-only aggregation and bootstrap scripts read immutable result tables. Hardware identifiers are provenance fields and are not treated as performance outcomes.

Table 22: Key integrity anchors. Full values and per-artifact manifests are included in the release.

Artifact class	Integrity anchor
WeedsGalore experiment archive	SHA-256 5d8395c274d990cd8fbfa728445f65c398be2b5d777bd5145eb9ae5ca4d9b68c
CWFID upstream source	Git commit 176612f3506c0ce70d85518cb82ddd938a6206c8
Locked CWFID protocol	SHA-256 d9c7bba0d9c6311551449bf44e6b7a0d50bbd3e4e63327a50b40f9eb9c7d1347
Strong-baseline configuration	SHA-256 d8007d26d5d49d330a2da8b3eb39670701e48943c0f22c4f43a1be217f9fdd93
Remote experiment backup	SHA-256 c42b4e3628face38cd4e2f654be146f6a33eb7353552f617392db84220375e95
Two-operator aggregate summary	SHA-256 233157c59d062b5d5577dcb12890089f87178e8527627633dcf1ec296b331fb9

The clean-room replay verifies configuration hashes before execution, reconstructs aggregate tables from archived predictions and exact matching outputs, compares regenerated source data with release checksums, and returns non-zero status on any mismatch. It also verifies that no credential, internal host, or private data field occurs in the public package.

Table 23: Preserved execution failures and resolution boundaries.

Stage	Failure	Resolution
CWFID acquisition	Several partial public downloads lacked upstream blobs	A separate complete candidate passed all 242 upstream blob checks; partial attempts remain archived
DeepLabV3+ initialisation	Network-dependent model construction was unavailable	Public weights were acquired once, hashed, and loaded locally without changing the frozen protocol
Semantic-ranking aggregation	One attempt encountered an existing output directory; another used an incorrect checkpoint root	New versioned outputs were created; failed attempts were retained
Locked CWFID evaluation	Empty candidate output was rejected by the table writer	Existing per-image outputs were consolidated without data re-opening or inference repetition
Remote backup transfer	The first large transfer was truncated	A second archive passed gzip, tar, extraction, entry-count, and SHA-256 verification; the truncated object remains marked invalid